Inductie structurala

n → n+1 -> inductie clasica, pe mutimea N

Inductie structurala -> pe tipuri de date abstracte (lista, arbore)
-> TDA se pot gandi ca fiind construite recursiv

T.D.A – tipuri de date abstracte

constructori
nulari – fara parametri -> se obtine []
externi – parametri din afara tipului de date -> se obtine [x]
interni – parametri din acelasi T.D.A (recursiv) -> se obtine [x] : [L] = [x L]

Tlist:
nil : → Tlist
//ext: T -> TList
cons: T x Tlist → Tlist

nil → intoarce o lista vida
cons → adauga T la inceput de lista

cons(1,[2,3,4]) = [1,2,3,4]
nil → []
//ext(3) -> [3]

head : TList\{nil}) → T
tail :TList\{nil} → Tlist
null:Tlist → {0,1}

head (cons (x,L)) = x
tail (cons (x,L)) = L
null(nil) = 1
null(cons(x,L)) = 0

concat (l1, l2) = l1 == nil? l2 : cons(head(l1),concat(tail(l1),l2)

```
concat(l1,l2){
        if (l1 == nil) //  C1
                return l2;
        else if (l1 == cons (y, z)) //C2
                return cons(y, concat(z,l2));
}
```

concat ([1,2,3],[4,5,6]) = cons(1,concat([2,3],[4,5,6]) = cons(1,cons(2,concat([3],[4,5,6])))=
cons(1,cons(2,cons(3,concat([], [4,5,6])))) = cons(1,cons(2,cons(3,[4,5,6])))
= cons(1,cons(2,[3,4,5,6])) = cons(1,[2,3,4,5,6]) = [1,2,3,4,5,6]

```
rev(l1) {
        if (l1 == nil) //R1
                return nil;
        else if (l1 == cons (x, l2)) //R2
                return concat( rev(l2),cons(x,nil))
        }
```

rev(l1) = l1 == nil? nil : concat(rev(tail(l1),cons(head(l1),nil))

rev([1,2,3,4,5,6]) = concat(rev([2,3,4,5,6]), cons(1,[])) = concat([6,5,4,3,2], cons(1,[]) = [6,5,4,3,2,1]

length(l1) = l1 == nil? 0 : 1+ length(tail(l1));

```
length(l1)
{
        if (l1 == nil)  //L1
                return 0;
        else if (l1 == cons(x,l2)) //L2
                return 1 + length(l2));
}
```

1. oricare l1 = Tlist, concat(l1,nil) = l1

Inductie structurala dupa l1.
Caz de baza:
l1 = nil C1
concat(l1,nil) = concat(nil,nil) = nil = l1

Ipoteza de inductie:
l1 = cons(x,l), concat(l,nil) = l
C2 ip.ind
concat(l1,nil) = concat(cons(x,l),nil) = cons(x, concat(l,nil)) = cons(x,l) = l1

2. l1,l2,l3 de tip Tlist, concat(l1,concat(l2,l3)) = concat(concat(l1,l2),l3)

Inductie dupa l1.
Caz de baza:
l1 = nil

C1
concat(l1,concat(l2,l3)) = concat(nil, concat(l2,l3)) = concat(l2,l3)
C1
concat(concat(l1,l2),l3) = concat(concat(nil,l2),l3) = concat(l2,l3)

Ip. de inductie:
l1 = cons(x,l) si presunem ca concat(l, concat(l2,l3)) = concat(concat(l,l2),l3)

C2 ip.ind
concat(l1,concat(l2,l3)) = concat(cons(x,l), concat(l2,l3)) = cons(x,concat(l,concat(l2,l3)) =
C2 C2
cons(x,concat(concat(l,l2),l3)) = concat(cons(x,concat(l,l2)), l3) =

concat(concat(cons(x,l),l2),l3) = concat (concat(l1,l2), l3)

3. l1,l2 de tip Tlist, rev(concat(l1,l2)) = concat(rev(l2),rev(l1))

Inductie dupa l1.
Caz de baza:
l1 = nil => C1
rev(concat(l1,l2)) = rev(concat(nil,l2)) = rev(l2)

R1 ex. 1
concat(rev(l2),rev(l1)) = concat(rev(l2),rev(nil)) = concat(rev(l2),nil) = rev(l2)

Ip. de inductie:
l1 = cons(x,l) si consideram adevarat rev(concat(l,l2)) = concat(rev(l2),rev(l))

C2 R2
rev(concat(l1,l2)) = rev(concat(cons(x,l),l2) = rev(cons(x,concat(l,l2)) =
ip.ind. ex. 2
concat(rev(concat(l,l2)), cons(x,nil)) = concat(concat(rev(l2),rev(l)), cons(x,nil)) =
R2
concat(rev(l2), concat(rev(l), cons(x,nil)) = concat(rev(l2),rev(cons(x,l)) =

concat(rev(l2), rev(l1))

4. length(concat(l1,l2)) = length(l1) + length(l2)

Inductie dupa l1.
Caz de baza:
l1 = nil C1 L1
length(concat(l1,l2)) = length(concat(nil,l2)) = length(l2) = 0 + length(l2) = length(nil) + length(l2)

= length(l1) + length(l2)

Ip. de inductie:
Presupunem ca l1 = cons(x,l) si consideram ca length(concat(l,l2)) = length(l) + length(l2)
C2 L2
length(concat(l1,l2)) = length(concat(cons(x,l),l2)) = length(cons(x,concat(l,l2)) =
ip. ind L2
1 + length(concat(l,l2)) = 1 + length(l) + length(l2) = length(cons(x,l)) + length(l2) =

length(l1) + length(l2)

5. length(rev(l)) = length(l)

Inductie dupa l.
Caz de baza: l = nil
R1
length(rev(l)) = length(rev(nil)) = length(nil) = length(l)

Ipoteza de inductie:
consideram ca l1 = cons(x,l) si length(rev(l)) = length(l)
R2 ex.4
length(rev(l1)) = length(rev(cons(x,l)) = length(concat(rev(l),cons(x,nil))) =
ip.ind, L2 L2, L1
length(rev(l)) + length(cons(x,nil)) = length(l) + 1 + length(nil) =

length(cons(x,l)) + 0 = length(l1)

C1: concat(nil, l2) = l2
C2: concat(cons(x,l1),l2) = cons(x,concat(l1,l2))
R1: rev(nil) = nil
R2: rev(cons(x,l)) = concat(rev(l),cons(x,nil))
L1: length(nil) = nil
L2: length(cons(x,l)) = 1 + length(l)

6. oricare l1 de tip Tlist, rev(rev(l1)) = l1

Inductie structurala dupa l1.
Caz de baza:
l1 = nil

$$rev(rev(l1)) = rev(rev(nil)) \overset{R1}{=} rev(nil) \overset{R1}{=} nil = l1$$

Ip. de inductie:

consideram l1 = cons(x,l) si rev(rev(l)) = l

$$rev(rev(l1)) = rev(rev(cons(x,l)) \overset{R2}{=} rev(concat(rev(l),cons(x,nil)) \overset{ex.\ 3}{=}$$
$$concat(rev(cons(x,nil)), rev(rev(l))) \overset{ip.ind}{=} concat(rev(cons(x,nil)), l) \overset{R2}{=}$$
$$concat(concat(rev(nil),cons(x,nil)),l) \overset{R1}{=} concat(concat(nil, cons(x,nil)),l) \overset{C1}{=}$$
$$concat(cons(x,nil),l) \overset{C2}{=} cons(x,concat(nil,l)) \overset{C1}{=} cons(x,l) = l1$$

fie urmatorii constructori:

a. : -> Tlist
b. TlistxTxTlist -> Tlist
c. T x T x Tlist -> Tlist
d. T x T x T -> Tlist

care din urmatorii constructori sunt de tip intern?
1. a si b
2. a si c
3. a si c si d
4. b si c si d
5. niciuna din variantele de mai sus